

24. Draw Line Charts of Alphabet Inc. Financial Data

Aim: To write a Python program to draw line charts of the financial data of Alphabet Inc. between October 3, 2016 and October 7, 2016 using data from a CSV file.

Algorithm:

1. Start the program.
2. Import Pandas and Matplotlib libraries.
3. Create a CSV file containing the financial data.
4. Read the CSV file using Pandas.
5. Extract the Date and Close columns.
6. Plot a line chart using Date on the x-axis and Close price on the y-axis.
7. Add x-axis label, y-axis label, and title.
8. Display the chart.
9. Stop the program.

Code:

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Create fdata.csv
```

```
with open("fdata.csv", "w") as f:
```

```
    f.write("Date,Open,High,Low,Close\n")
```

```
    f.write("10-03-16,774.25,776.065002,769.5,772.559998\n")
```

```
    f.write("10-04-16,776.030029,778.710022,772.890015,776.429993\n")
```

```
    f.write("10-05-16,779.309998,782.070007,775.650024,776.469971\n")
```

```
    f.write("10-06-16,779,780.47998,775.539978,776.859985\n")
```

```
    f.write("10-07-16,779.659973,779.659973,770.75,775.080017\n")
```

```
# Read CSV file
```

```
df = pd.read_csv("fdata.csv")
```

Plot all financial data

```
plt.plot(df["Date"], df["Open"], label="Open")
```

```
plt.plot(df["Date"], df["High"], label="High")
```

```
plt.plot(df["Date"], df["Low"], label="Low")
```

```
plt.plot(df["Date"], df["Close"], label="Close")
```

```
plt.xlabel("Date")
```

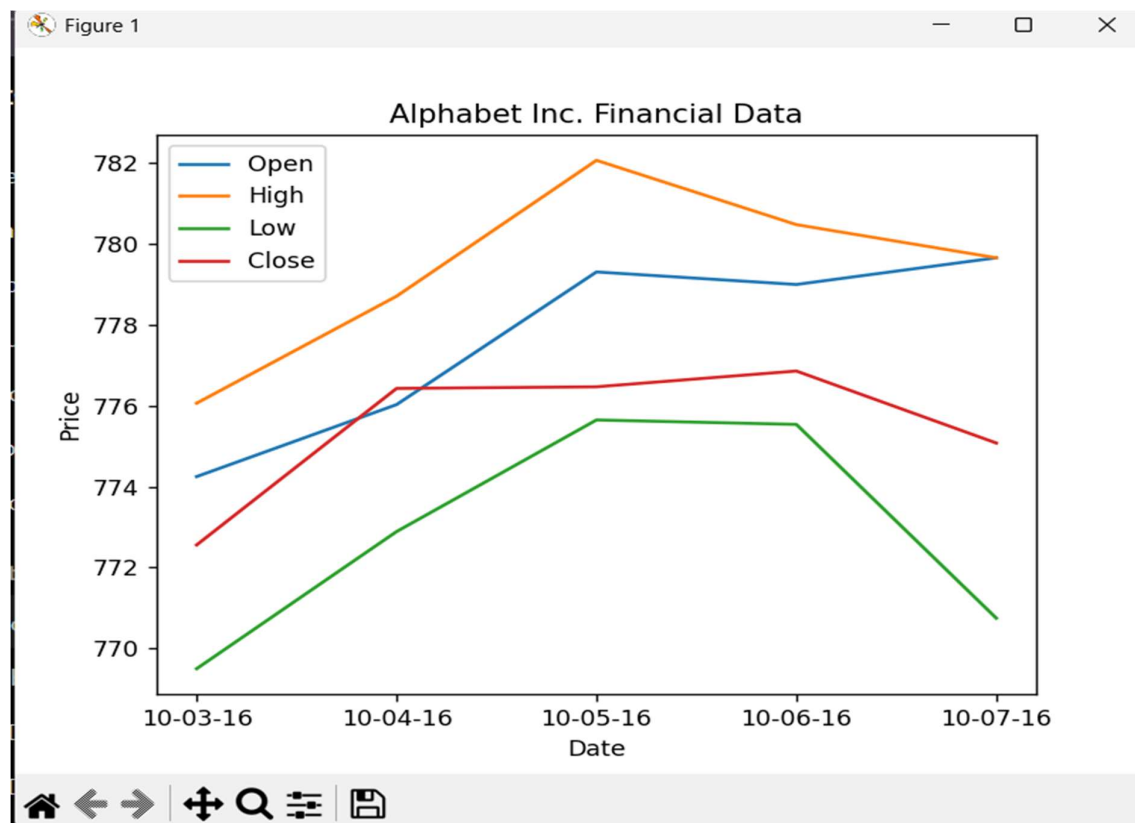
```
plt.ylabel("Price")
```

```
plt.title("Alphabet Inc. Financial Data")
```

```
plt.legend()
```

```
plt.show()
```

OUTPUT:



25. Plot Two or More Lines with Legends, Different Widths and Colors

Aim: To write a Python program to plot two or more lines with legends, different colors, and line widths.

Algorithm:

1. Start the program.
2. Import matplotlib.pyplot.
3. Create x-axis and y-axis values for two lines.
4. Plot the first line with a specified color and width.
5. Plot the second line with a different color and width.
6. Add labels for x-axis and y-axis.
7. Add title and legend.
8. Display the graph.
9. Stop the program.

Code:

```
import matplotlib.pyplot as plt
```

```
x = [10, 20, 30]
```

```
y1 = [20, 40, 10]
```

```
y2 = [40, 10, 30]
```

```
plt.plot(x, y1, color='blue', linewidth=3, label='line1-width-3')
```

```
plt.plot(x, y2, color='red', linewidth=5, label='line2-width-5')
```

```
plt.xlabel('x-axis')
```

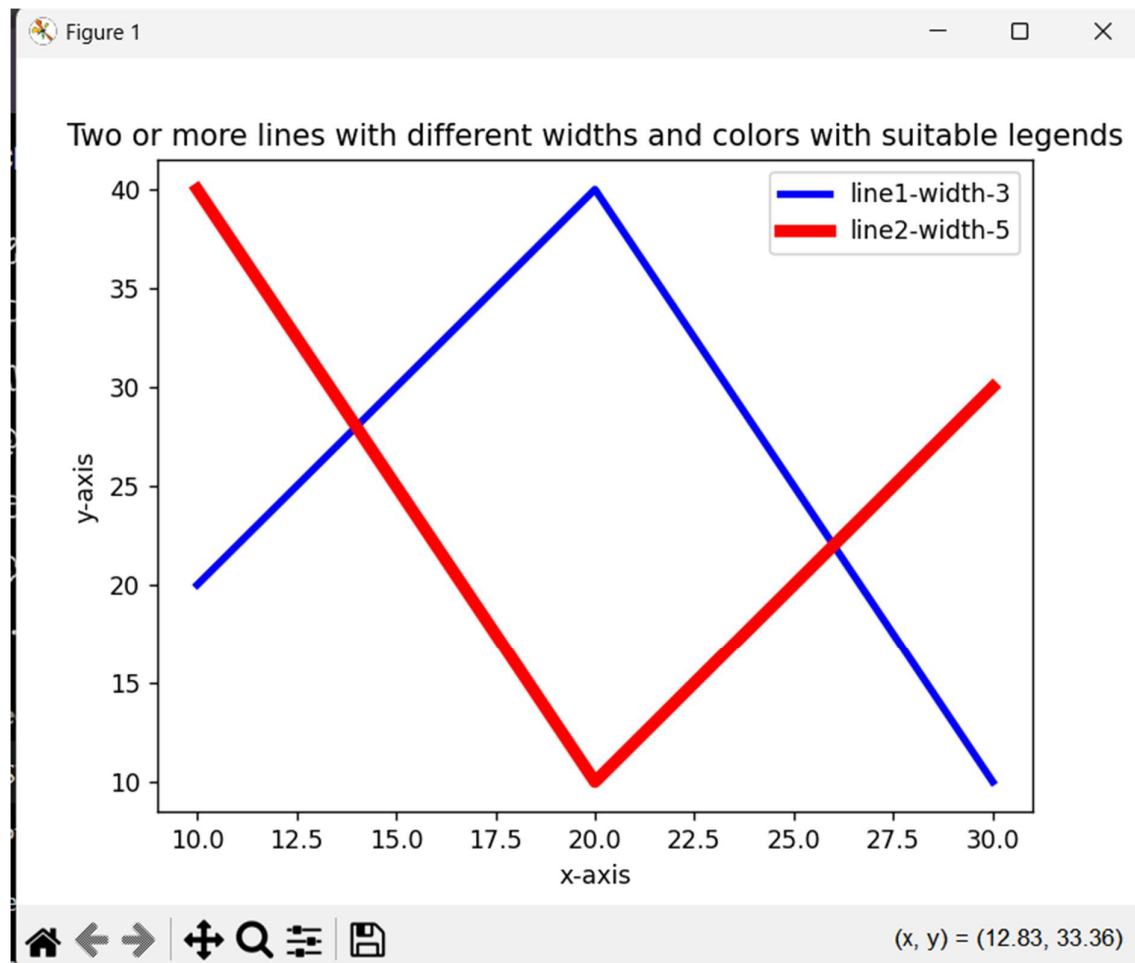
```
plt.ylabel('y-axis')
```

```
plt.title('Two or more lines with different widths and colors with suitable legends')
```

```
plt.legend()
```

```
plt.show()
```

OUTPUT:



26. Create Multiple Plots

Aim: To write a Python program to create multiple plots using Matplotlib.

Algorithm:

1. Start the program.
2. Import matplotlib.pyplot.
3. Create a figure.
4. Divide the figure into multiple subplots using subplot().
5. Plot data in each subplot.
6. Display all plots using show().
7. Stop the program.

Code:

```
import matplotlib.pyplot as plt
```

```
plt.subplot(2, 1, 1)
```

```
plt.plot([1, 2, 3, 4])
```

```
plt.subplot(2, 3, 4)
```

```
plt.plot([1, 2, 3])
```

```
plt.subplot(2, 3, 5)
```

```
plt.plot([3, 2, 1])
```

```
plt.subplot(2, 3, 6)
```

```
plt.plot([1, 3, 2])
```

```
plt.show()
```

OUTPUT:

